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ACT SUBJECT TEST — MATH 2 • UNIT 3.01

# Sequences, Series & Binomial Theorem

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

**Total Questions:** 25 • Verified Step-by-Step Solutions

## QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

| | | | | |
|---|---|---|---|---|
| Q1 B | Q2 A | Q3 A | Q4 B | Q5 C |
| Q6 A | Q7 B | Q8 C | Q9 D | Q10 D |
| Q11 B | Q12 D | Q13 C | Q14 A | Q15 D |
| Q16 D | Q17 B | Q18 B | Q19 A | Q20 A |
| Q21 A | Q22 C | Q23 C | Q24 B | Q25 C |

## Sequences, Series & Binomial Theorem Practice Exercises

**1** MCQ

The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?

A 131

B 138 ✓

C 140

D 145

### WORKED SOLUTION & VERIFICATION

To find the 20th term of the arithmetic sequence, first identify the initial term and the common difference:

1. The first term is $a_1 = 5$.

2. The common difference $d$ is found by subtracting consecutive terms: $d = 12 - 5 = 7$.

The general formula for the $n$th term of an arithmetic sequence is:

$$a_n = a_1 + (n-1)d$$

Substitute $n = 20$, $a_1 = 5$, and $d = 7$:

$$a_{20} = 5 + (20-1)(7) = 5 + 19(7) = 5 + 133 = 138$$

Common misconceptions:

- Using $n$ instead of $n-1$ yields $5 + 20(7) = 145$.
- Calculating $20 \times 7 = 140$ overlooks the starting value $a_1 = 5$.
- Using $n-2$ steps yields $5 + 18(7) = 131$.

**2** MCQ

In an arithmetic sequence, the 3rd term is $k + 7$, the 6th term is $3k - 1$, and the 11th term is $6k - 8$, where $k$ is a real constant. What is the first term, $a_1$, of this sequence?

A $6$ ✓

B $10$

C $19$

D $26$

**WORKED SOLUTION & VERIFICATION**

In an arithmetic sequence with first term $a_1$ and common difference $d$, the $n$th term is given by $a_n = a_1 + (n-1)d$.

Step 1: Express the common difference $d$ in terms of $k$ using the differences between the given terms:

$a_6 - a_3 = 3d = (3k - 1) - (k + 7) = 2k - 8 \implies d = \frac{2k-8}{3}$

$a_{11} - a_6 = 5d = (6k - 8) - (3k - 1) = 3k - 7 \implies d = \frac{3k-7}{5}$

Step 2: Equate the two expressions for $d$ to solve for $k$:

$\frac{2k-8}{3} = \frac{3k-7}{5}$

$5(2k - 8) = 3(3k - 7)$

$10k - 40 = 9k - 21$

$k = 19$

Step 3: Determine $d$ and $a_3$:

$d = \frac{2(19)-8}{3} = \frac{30}{3} = 10$

$a_3 = 19 + 7 = 26$

Step 4: Find the first term $a_1$:

$a_1 = a_3 - 2d = 26 - 2(10) = 6$

Common traps:

- Selecting $19$ confuses the parameter $k$ with the first term.
- Selecting $10$ stops at finding the common difference $d$.
- Selecting $26$ stops at the value of the 3rd term $a_3$.

**3** MCQ

An arithmetic sequence $(a_n)$ and a geometric sequence $(b_n)$ of non-zero real numbers share the same first term, so $a_1 = b_1$. The sequences also satisfy $a_5 = b_2$ and $a_6 = b_3$, with $b_2 \neq b_1$. If the maximum value of the partial sum $S_N = \sum_{k=1}^{N} a_k$ of the arithmetic sequence is $153$, what is the sum of the infinite geometric series $\sum_{n=1}^{\infty} b_n$?

- **A** 64 ✓
- **B** 96
- **C** 144
- **D** 192

**WORKED SOLUTION & VERIFICATION**

Let $a_1 = b_1 = a$, let $d$ be the common difference of $(a_n)$, and let $r$ be the common ratio of $(b_n)$.

1. Express the given terms:

$$a_5 = a + 4d = b_2 = ar$$

$$a_6 = a + 5d = b_3 = ar^2$$

Subtracting the first equation from the second gives:

$$d = a_6 - a_5 = ar^2 - ar = ar(r-1)$$

Substitute $d$ back into $a + 4d = ar$:

$$a + 4ar(r-1) = ar$$

Since the terms of the sequence are non-zero, $a \neq 0$. Dividing by $a$:

$$1 + 4r(r-1) = r \implies 4r^2 - 5r + 1 = 0$$

$$(4r-1)(r-1) = 0$$

Since $b_2 \neq b_1$, $r \neq 1$. Therefore, $r = \frac{1}{4}$.

2. Determine $d$ in terms of $a$:

$$d = ar(r-1) = a\left(\frac{1}{4}\right)\left(-\frac{3}{4}\right) = -\frac{3}{16}a$$

3. Optimize the partial sum $S_N$:

For $S_N$ to achieve a finite maximum, the common difference must be negative, so $d < 0 \implies a > 0$.

The $n$-th term of the arithmetic sequence is:

$$a_n = a + (n-1)\left(-\frac{3}{16}a\right) = \frac{a}{16}(19 - 3n)$$

Evaluating the sign of $a_n$:

- For $n \leq 6$, $19 - 3n > 0 \implies a_n > 0$
- For $n \geq 7$, $19 - 3n < 0 \implies a_n < 0$

Because terms are added until they become negative, $S_N$ attains its unique maximum at $N = 6$.

4. Solve for $a$:

$$S_6 = \frac{6}{2}(a_1 + a_6) = 3\left(a + \frac{a}{16}\right) = 3 \cdot \frac{17a}{16} = \frac{51a}{16}$$

Given $S_6 = 153$:

$$\frac{51a}{16} = 153 \implies a = 153 \cdot \frac{16}{51} = 3 \cdot 16 = 48$$

5. Sum of the infinite geometric series:
Since $|r| = \frac{1}{4} < 1$, the infinite series converges to:

$$\sum_{n=1}^{\infty} b_n = \frac{b_1}{1-r} = \frac{48}{1-\frac{1}{4}} = \frac{48}{\frac{3}{4}} = 64$$

**4** MCQ

The bar chart below shows the values of the first four terms of an arithmetic sequence, $a_1, a_2, a_3$, and $a_4$, where $a_1 = 6$, $a_2 = 11$, $a_3 = 16$, and $a_4 = 21$. What is the value of the 8th term, $a_8$, of this sequence?

A 36

B **41** ✓

C 46

D 48

**WORKED SOLUTION & VERIFICATION**

1. Find the common difference $d$ by subtracting consecutive terms:

$$d = a_2 - a_1 = 11 - 6 = 5$$

Check consistency with subsequent terms: $16 - 11 = 5$ and $21 - 16 = 5$.
2. Use the standard formula for the $n$th term of an arithmetic sequence:

$$a_n = a_1 + (n-1)d$$

For the 8th term ($n = 8$):

$$a_8 = 6 + (8-1)(5) = 6 + 7(5) = 6 + 35 = 41$$

Alternatively, continuing term-by-term: $a_5 = 26$, $a_6 = 31$, $a_7 = 36$, $a_8 = 41$.
Common Errors:
- Choosing 36 corresponds to finding the 7th term $a_7$ due to an off-by-one error.
- Choosing 46 results from using $a_1 + 8d$ instead of $a_1 + (8-1)d$.
- Choosing 48 results from incorrectly treating the sequence as direct multiplication, $6 \times 8$.

**5** MCQ

The first three terms of a geometric sequence of positive real numbers are $k-1$, $k+2$, and $3k$, in that order. What is the value of the 6th term of this sequence?

A 24

B 48

C 96 ✓

D 192

**WORKED SOLUTION & VERIFICATION**

Step 1: Set up the ratio property for consecutive terms of a geometric sequence.
Since the sequence is geometric, the common ratio $r$ satisfies:

$$r = \frac{a_2}{a_1} = \frac{a_3}{a_2} \implies \frac{k+2}{k-1} = \frac{3k}{k+2}$$

Step 2: Solve the quadratic equation for $k$.
Cross-multiplying gives:

$$(k+2)^2 = 3k(k-1)$$

$$k^2 + 4k + 4 = 3k^2 - 3k$$

$$2k^2 - 7k - 4 = 0$$

Factoring the quadratic:

$$(2k+1)(k-4) = 0$$

Thus, $k = -\frac{1}{2}$ or $k = 4$.
Step 3: Discard the extraneous solution.
The problem states that all terms in the sequence are positive real numbers. If $k = -\frac{1}{2}$, the first term is $k-1 = -\frac{3}{2} < 0$, which contradicts the condition. Therefore, $k = 4$.
Step 4: Find the first term and common ratio.
- First term: $a_1 = 4 - 1 = 3$
- Second term: $a_2 = 4 + 2 = 6$
- Common ratio: $r = \frac{6}{3} = 2$

Step 5: Compute the 6th term.
Using the formula for the $n$th term of a geometric sequence, $a_n = a_1 r^{n-1}$:

$$a_6 = 3 \cdot 2^{6-1} = 3 \cdot 2^5 = 3 \cdot 32 = 96$$

Misconception Traps:
- Choice A (24) is the 4th term ($3 \cdot 2^3$).
- Choice B (48) is the 5th term ($3 \cdot 2^4$), resulting from an off-by-one exponent error.
- Choice D (192) incorrectly uses $a_n = a_1 r^n$ giving $3 \cdot 2^6 = 192$.

**6** MCQ

An infinite geometric sequence of positive real numbers has first term $a_1$ and common ratio $r \in (0, 1)$. Let $S_1 = \sum_{n=1}^{\infty} a_n$, $S_2 = \sum_{n=1}^{\infty} a_n^2$, and $S_3 = \sum_{n=1}^{\infty} a_n^3$. If $S_1 = 4$ and $S_2 = 8$, what is the value of $S_3$?

- **A** $\frac{256}{13}$ ✓
- **B** $\frac{128}{13}$
- **C** $\frac{64}{7}$
- **D** $16$

**WORKED SOLUTION & VERIFICATION**

For an infinite geometric sequence with first term $a_1 > 0$ and ratio $r \in (0, 1)$, the terms are $a_n = a_1 r^{n-1}$.

1. The sum of the original sequence is:

$$S_1 = \frac{a_1}{1-r} = 4 \implies a_1 = 4(1-r)$$

2. The terms of the squared sequence are $a_n^2 = a_1^2 (r^2)^{n-1}$, which forms a geometric sequence with first term $a_1^2$ and common ratio $r^2$. Its sum is:

$$S_2 = \frac{a_1^2}{1-r^2} = 8$$

3. Relate $S_2$ and $S_1$ using the algebraic identity $1 - r^2 = (1-r)(1+r)$:

$$\frac{S_2}{S_1^2} = \frac{\frac{a_1^2}{1-r^2}}{\left(\frac{a_1}{1-r}\right)^2} = \frac{(1-r)^2}{1-r^2} = \frac{1-r}{1+r}$$

Substituting the known numerical values $S_1 = 4$ and $S_2 = 8$:

$$\frac{8}{4^2} = \frac{8}{16} = \frac{1}{2} \implies \frac{1-r}{1+r} = \frac{1}{2}$$

Cross-multiplying gives:

$$2(1-r) = 1+r \implies 2-2r = 1+r \implies 3r = 1 \implies r = \frac{1}{3}$$

4. Find the first term $a_1$:

$$a_1 = 4\left(1 - \frac{1}{3}\right) = 4 \cdot \frac{2}{3} = \frac{8}{3}$$

5. The cubed sequence has terms $a_n^3 = a_1^3 (r^3)^{n-1}$, with first term $a_1^3$ and ratio $r^3$:

$$S_3 = \frac{a_1^3}{1-r^3} = \frac{\left(\frac{8}{3}\right)^3}{1-\left(\frac{1}{3}\right)^3} = \frac{\frac{512}{27}}{1-\frac{1}{27}} = \frac{\frac{512}{27}}{\frac{26}{27}} = \frac{512}{26} = \frac{256}{13}$$

Common misconception: Dividing $S_2$ directly by $S_1$ to get $\frac{a_1}{1+r} = 2$ and incorrectly assuming $r = 1/2$ leads to $\frac{64}{7}$.

**7** MCQ

The first four terms of an arithmetic sequence are shown below.

$$4, 11, 18, 25, \ldots$$

What is the 31st term of the sequence?

A 210

B **214** ✓

C 217

D 221

**WORKED SOLUTION & VERIFICATION**

To find the 31st term of an arithmetic sequence, use the formula for the $n$th term:

$$a_n = a_1 + (n-1)d$$

1. Identify the first term $a_1$:

$$a_1 = 4$$

2. Find the common difference $d$ by subtracting consecutive terms:

$$d = 11 - 4 = 7$$

3. Substitute $n = 31$, $a_1 = 4$, and $d = 7$ into the formula:

$$a_{31} = 4 + (31-1)(7) = 4 + 30(7) = 4 + 210 = 214$$

- A common error is using $n$ instead of $n-1$, which yields $4 + 31(7) = 221$.
- Another common error is computing $31 \times 7 = 217$ or forgetting to add the initial term $a_1 = 4$, giving $210$.

**8** MCQ

The second term of a geometric sequence of positive real numbers is $k$, and the fifth term is $k^2 - 20$. If the common ratio of the sequence is $2$, what is the sum of the first $6$ terms of the sequence?

A 155

B 160

C 315 ✓

D 630

**WORKED SOLUTION & VERIFICATION**

Step 1: Express the terms in terms of the first term $a_1$ and common ratio $r = 2$.

The $n$-th term of a geometric sequence is given by $a_n = a_1 r^{n-1}$.

For the second term:

$$a_2 = a_1(2)^{2-1} = 2a_1 = k \implies a_1 = \frac{k}{2}$$

For the fifth term:

$$a_5 = a_1(2)^{5-1} = 16a_1$$

Substituting $a_1 = \frac{k}{2}$ gives:

$$a_5 = 16\left(\frac{k}{2}\right) = 8k$$

Step 2: Solve for the parameter $k$.

We are given $a_5 = k^2 - 20$, so:

$$k^2 - 20 = 8k \implies k^2 - 8k - 20 = 0$$

Factoring the quadratic equation:

$$(k - 10)(k + 2) = 0$$

Since the sequence consists of positive real numbers, $k > 0$, so $k = 10$.

Step 3: Determine $a_1$ and compute the sum $S_6$.

The first term is $a_1 = \frac{10}{2} = 5$.

The sum of the first $n$ terms of a geometric sequence is given by $S_n = \frac{a_1(r^n - 1)}{r - 1}$.

For $n = 6$:

$$S_6 = \frac{5(2^6 - 1)}{2 - 1} = 5(64 - 1) = 5(63) = 315$$

Common traps:

- Confusing the sum with the 6th term gives $a_6 = 5(2^5) = 160$.
- Summing only 5 terms gives $S_5 = 5(31) = 155$.
- Forgetting to divide $k$ by $2$ to find $a_1$ (mistaking $a_2$ for $a_1$) gives $10(63) = 630$.

**9** MCQ

An infinite geometric sequence $a_1, a_2, a_3, \ldots$ of positive real numbers has common ratio $r < 1$ and sum $S$. The sub-sequence of terms whose indices are of the form $3k - 1$ for positive integers $k$ (that is, $a_2, a_5, a_8, \ldots$) has sum

$$\sum_{k=1}^{\infty} a_{3k-1} = \frac{3}{13}S.$$

Furthermore, the infinite series formed by the squares of the terms has sum

$$\sum_{n=1}^{\infty} a_n^2 = \frac{9}{2}S.$$

What is the value of $\sum_{n=1}^{\infty} na_n$?

A $\frac{9}{2}$

B $\frac{27}{4}$

C $\frac{81}{4}$

D $\frac{27}{2}$ ✓

**WORKED SOLUTION & VERIFICATION**

Let the first term be $a_1 = a > 0$ and the common ratio be $r \in (0, 1)$. The sum of the original series is:

$$S = \sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}.$$

**Step 1: Determine the common ratio $r$.**

The sub-sequence consists of terms $a_2, a_5, a_8, \ldots$, which has first term $a_2 = ar$ and common ratio $r^3$. Its sum is:

$$\sum_{k=1}^{\infty} a_{3k-1} = \frac{ar}{1-r^3}.$$

Given that this sum is $\frac{3}{13}S$:

$$\frac{ar}{1-r^3} = \frac{3}{13} \cdot \frac{a}{1-r}.$$

Since $a > 0$ and $1 - r^3 = (1-r)(1+r+r^2)$, we divide both sides by $\frac{a}{1-r}$:

$$\frac{r}{1+r+r^2} = \frac{3}{13} \implies 13r = 3(1+r+r^2) \implies 3r^2 - 10r + 3 = 0.$$

Factoring gives $(3r-1)(r-3) = 0$. Since $r < 1$, we have $r = \frac{1}{3}$.

**Step 2: Determine the first term $a$.**

The series of squares has first term $a^2$ and common ratio $r^2$:

$$\sum_{n=1}^{\infty} a_n^2 = \frac{a^2}{1-r^2}.$$

Given $\sum_{n=1}^{\infty} a_n^2 = \frac{9}{2}S$:

$$\frac{a^2}{(1-r)(1+r)} = \frac{9}{2} \cdot \frac{a}{1-r} \implies \frac{a}{1+r} = \frac{9}{2}.$$

Substituting $r = \frac{1}{3}$:

$$a = \frac{9}{2}\left(1 + \frac{1}{3}\right) = \frac{9}{2} \cdot \frac{4}{3} = 6.$$

**Step 3: Evaluate the arithmetico-geometric series.**
We seek $T = \sum_{n=1}^{\infty} na_n = 6\sum_{n=1}^{\infty} n\left(\frac{1}{3}\right)^{n-1}$.
Let $U = \sum_{n=1}^{\infty} nr^{n-1} = 1 + 2r + 3r^2 + \ldots$. Multiplying by $r$ and subtracting:

$$U - rU = 1 + r + r^2 + \cdots = \frac{1}{1-r} \implies U = \frac{1}{(1-r)^2}.$$

For $r = \frac{1}{3}$:

$$U = \frac{1}{\left(1 - \frac{1}{3}\right)^2} = \frac{1}{\frac{4}{9}} = \frac{9}{4}.$$

Therefore,

$$T = 6 \cdot \frac{9}{4} = \frac{27}{2}.$$

**Common Misconceptions:**
- Choice B ($\frac{81}{4}$) mistakenly multiplies $U$ by the total series sum $S = 9$ instead of the first term $a = 6$.
- Choice C ($\frac{9}{2}$) uses the shifted index expression $\frac{ar}{(1-r)^2}$.
- Choice D ($\frac{27}{4}$) incorrectly uses $1 - r^2$ in the denominator instead of $(1-r)^2$.

**10** MCQ

In an arithmetic sequence, the first term is $a_1 = 4$ and the fifth term is $a_5 = 28$. What is the value of the tenth term, $a_{10}$?

A 52
B 56
C 64
D **58** ✓

**WORKED SOLUTION & VERIFICATION**

The $n$th term of an arithmetic sequence is given by the formula:

$$a_n = a_1 + (n-1)d$$

where $a_1$ is the first term and $d$ is the common difference.
Using the fifth term $a_5 = 28$:

$$28 = 4 + (5-1)d \implies 28 = 4 + 4d \implies 4d = 24 \implies d = 6$$

Now, calculate the tenth term $a_{10}$:

$$a_{10} = a_1 + (10-1)d = 4 + 9(6) = 4 + 54 = 58$$

Common misconceptions:
- Choosing 56 incorrectly assumes direct proportionality (e.g., $a_{10} = 2 \times a_5 = 2 \times 28$), which does not hold for arithmetic sequences with a non-zero initial offset.
- Choosing 64 results from using the formula $a_n = a_1 + nd$ instead of $a_1 + (n-1)d$, giving $4 + 10(6) = 64$.
- Choosing 52 results from dividing the difference by 5 instead of 4 to find $d$ ($d = 4.8$), then evaluating $4 + 10(4.8) = 52$.

**11** MCQ

A geometric sequence has $a_1 = 3$ and common ratio $r = 3$. Find $a_3$.

| | |
|---|---|
| A 81 | **B 27** ✓ |
| C 30 | D -27 |

**WORKED SOLUTION & VERIFICATION**

Applying Sequences & Series principles, $a_3 = a_1 r^{n-1} = 3(3)^2 = 27$. Therefore, the verified correct option is 27.

**12** MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = -3$. Find $a_3$.

| | |
|---|---|
| A 51 | B -54 |
| C -162 | **D 54** ✓ |

**WORKED SOLUTION & VERIFICATION**

Applying Sequences & Series principles, $a_3 = a_1 r^{n-1} = 6(-3)^2 = 54$. Therefore, the verified correct option is 54.

**13** MCQ

A geometric sequence has $a_1 = 2$ and common ratio $r = -2$. Find $a_6$.

| | |
|---|---|
| A -66 | B 64 |
| **C -64** ✓ | D 128 |

**WORKED SOLUTION & VERIFICATION**

$a_6 = a_1 r^{n-1} = 2(-2)^5 = -64$.

**14** MCQ

A geometric sequence has $a_1 = 4$ and common ratio $r = 2$. Find $a_3$.

| | |
|---|---|
| **A 16** ✓ | B 32 |
| C 18 | D -16 |

**WORKED SOLUTION & VERIFICATION**

Applying Sequences & Series principles, $a_3 = a_1 r^{n-1} = 4(2)^2 = 16$. Therefore, the verified correct option is 16.

**15** MCQ

In an applied problem, a geometric sequence has $a_1 = 4$ and common ratio $r = -3$. Find $a_5$.

| | |
|---|---|
| A 321 | B -324 |
| C -972 | **D 324** ✓ |

**WORKED SOLUTION & VERIFICATION**

$a_5 = a_1 r^{n-1} = 4(-3)^4 = 324$.

**16** MCQ

Suppose a geometric sequence has $a_1 = 3$ and common ratio $r = -2$. Find $a_4$.

| | |
|---|---|
| A 24 | B 48 |
| C -26 | **D -24** ✓ |

**WORKED SOLUTION & VERIFICATION**

$a_4 = a_1 r^{n-1} = 3(-2)^3 = -24.$

**17** MCQ

An observed geometric sequence has $a_1 = 5$ and common ratio $r = -2$. Find $a_4$.

| | |
|---|---|
| A -42 | **B -40** ✓ |
| C 40 | D 80 |

**WORKED SOLUTION & VERIFICATION**

$a_4 = a_1 r^{n-1} = 5(-2)^3 = -40.$

**18** MCQ

Consider a geometric sequence has $a_1 = 5$ and common ratio $r = -3$. Find $a_5$.

| | |
|---|---|
| A 402 | **B 405** ✓ |
| C -405 | D -1215 |

**WORKED SOLUTION & VERIFICATION**

$a_5 = a_1 r^{n-1} = 5(-3)^4 = 405.$

**19** MCQ

In an applied problem, a geometric sequence has $a_1 = 3$ and common ratio $r = 2$. Find $a_4$.

| | |
|---|---|
| **A 24** ✓ | B 26 |
| C -24 | D 48 |

**WORKED SOLUTION & VERIFICATION**

Applying Sequences & Series principles, $a_4 = a_1 r^{n-1} = 3(2)^3 = 24$. Therefore, the verified correct option is 24.

**20** MCQ

A geometric sequence has $a_1 = 1$ and common ratio $r = -2$. Find $a_6$.

| | |
|---|---|
| **A -32** ✓ | B -34 |
| C 32 | D 64 |

**WORKED SOLUTION & VERIFICATION**

$a_6 = a_1 r^{n-1} = 1(-2)^5 = -32.$

**21** MCQ

In an applied problem, a geometric sequence has $a_1 = 1$ and common ratio $r = -2$. Find $a_5$.

- **A** 16 ✓
- **B** -32
- **C** 14
- **D** -16

**WORKED SOLUTION & VERIFICATION**

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 1(-2)^4 = 16$. Therefore, the verified correct option is 16.

**22** MCQ

Suppose a geometric sequence has $a_1 = 2$ and common ratio $r = 3$. Find $a_6$.

- **A** 489
- **B** -486
- **C** 486 ✓
- **D** 1458

**WORKED SOLUTION & VERIFICATION**

Applying Sequences & Series principles, $a_6 = a_1 r^{n-1} = 2(3)^5 = 486$. Therefore, the verified correct option is 486.

**23** MCQ

An observed geometric sequence has $a_1 = 2$ and common ratio $r = 3$. Find $a_3$.

- **A** -18
- **B** 54
- **C** 18 ✓
- **D** 21

**WORKED SOLUTION & VERIFICATION**

Applying Sequences & Series principles, $a_3 = a_1 r^{n-1} = 2(3)^2 = 18$. Therefore, the verified correct option is 18.

**24** MCQ

Consider a geometric sequence has $a_1 = 3$ and common ratio $r = 2$. Find $a_6$.

- **A** -96
- **B** 96 ✓
- **C** 192
- **D** 98

**WORKED SOLUTION & VERIFICATION**

Applying Sequences & Series principles, $a_6 = a_1 r^{n-1} = 3(2)^5 = 96$. Therefore, the verified correct option is 96.

**25** MCQ

Suppose a geometric sequence has $a_1 = 5$ and common ratio $r = -3$. Find $a_3$.

- **A** -135
- **B** 42
- **C** 45 ✓
- **D** -45

**WORKED SOLUTION & VERIFICATION**

Applying Sequences & Series principles, $a_3 = a_1 r^{n-1} = 5(-3)^2 = 45$. Therefore, the verified correct option is 45.